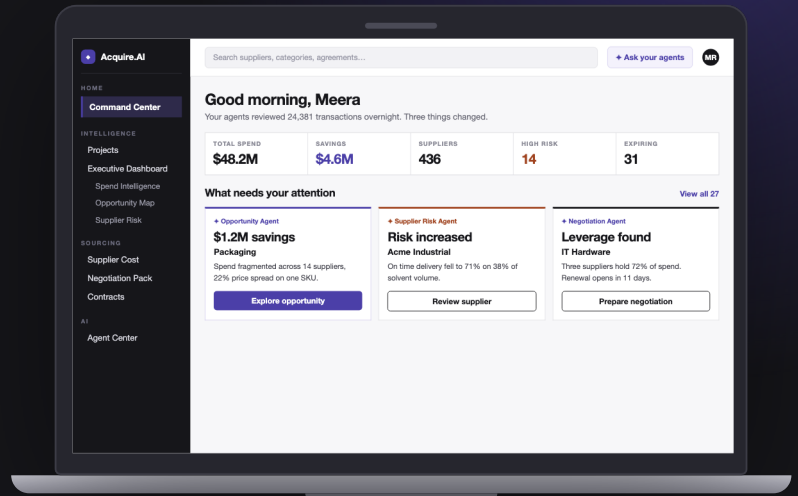


ENTERPRISE B2B · PROCUREMENT INTELLIGENCE

From 45 minutes to 11 minutes per agreement

Enterprise procurement rebuilt around five AI agents. The agents gather, compare and chase. The buyer only decides.



COMMAND CENTER · HIGH FIDELITY

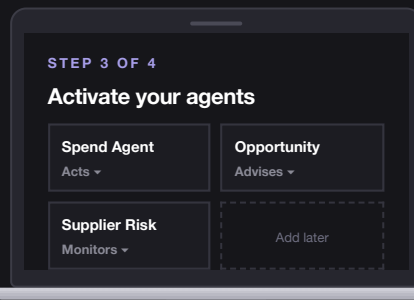
11 min per agreement, down from 45

5 agents replacing 39 minutes of clerical work

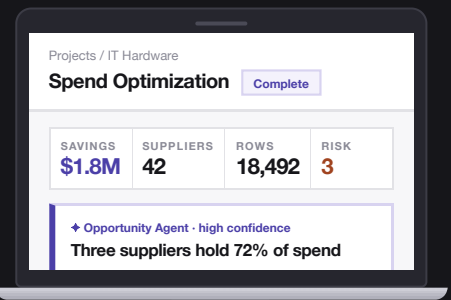
4 → 1 tools collapsed into one workspace



WIREFRAME · COMMAND CENTER



ONBOARDING · AGENT ACTIVATION



PROJECT RESULT · FINDINGS

ROLE

Lead UX designer. Research, IA, interaction design, agent behaviour, design system

WORKED WITH

Business, engineering, the AI/ML team, and senior leadership

SCOPE

Onboarding, command center, agent system, sourcing workspace

Every raw material a factory buys sits behind an agreement with an expiry date

A factory buys dozens of raw materials, and most materials can be bought from several vendors. Each supply relationship sits behind an agreement that fixes the cost per unit, the start and end dates, and who is responsible for getting the goods there.

Every agreement expires, so the same work comes back around. As an expiry gets close, the procurement team sends a fresh RFP to the qualified vendors,

collects their proposals, negotiates, and awards the next agreement.

A mid-size plant renews hundreds of these a year. The work repeats, the deadlines are fixed, and almost nothing helps. The buyer is the only thing holding email, spreadsheets and phone calls together.

A NOTE ON EVIDENCE

The project started from a real industry brief. Where primary research with buyers was not possible, this document says so and marks the reasoning as inferred. Numbers labelled △ modelled come from breaking the documented process into steps, not from measured data.

GLOSSARY

The whole product rests on seven words. Getting them right in the interface removed most of the confusion that showed up in early reviews.

RFP	Request for Proposal. The manufacturer's invitation to bid on a new agreement, with a deadline.
Product	A raw material. There is always more than one vendor capable of supplying it.
Product Master	The manufacturer's internal list of every product it buys. The spine of the data model.
Vendor	Supplies one or more products. Competes with other vendors for each agreement.
Agreement	Cost per unit, start date, end date, and delivery mode. Either the vendor delivers or the manufacturer collects.
Reverse auction	One buyer, many sellers, price moving down. The market structure I designed the bidding around.
Agent	A scoped, named piece of automation with one job, a stated confidence, and a visible audit trail.

Nine steps, four tools, one spreadsheet holding it together

Before any screens, the process was mapped end to end. Nine steps, and at every hand-off the work leaves the system of record and lands in someone’s inbox.

MANUFACTURER · PROCUREMENT TEAM

1

Pull every agreement expiring in the next 15 days

Excel filter, run manually

4

Compile all proposals received before the deadline

Copy-paste from mail into a sheet

7

Compile final bids from every vendor into one comparison

Second manual consolidation

8

Choose the most suitable offer

The only judgement step

VENDORS (8 TO 14 PER PRODUCT)

3

Reply to the mail with a proposal

Free-format, no schema

6

Send a final offer after negotiation

Blind to competing prices

VENDOR POOL

Vendor 1

Vendor 2

Vendor 3

Vendor 4

...

Vendor N

Step 2 →

RFP floated by email, with a deadline

Step 5 →

Counter price per unit, negotiated over mail or phone

Step 9 ↓

The accepted bid becomes the signed agreement

WHERE THE 45 MINUTES GOES △ MODELLED

Find expiring agreements	6'
Draft and send the RFP	8'
Chase and track replies	9'
Compile into a comparison sheet	12'
Negotiate over mail and phone	7'
Compare and select	3'

Three of the forty-five minutes are spent buying. The other forty-two are spent getting ready to buy.

That ratio, not the total, became the design target.

The brief asked for a faster process. The problem was that nobody owned the middle.

The brief came with a number in it. Read step by step, the number was measuring the wrong thing.

Digitize the procurement process and bring every stakeholder onto a single platform. The process currently takes around 45 minutes per agreement; reduce it to 15.

THE ORIGINAL BRIEF

THREE PROBLEMS, AND THE TRADE OFF EACH ANSWER COST

42' of clerical work accounted for Δ modelled

01 18 RETURNED TO THE BUYER	Nobody owns the record Prices sit in mail threads, history sits in Excel, terms sit in PDFs. Four tools per agreement, and no single answer to what a vendor has ever quoted.	WEIGHED AND DROPPED ✗ Wait for clean data before onboarding a customer ✗ Build a bespoke importer per source system	THE CALL, AND WHAT IT COST Ingest the mess, then show a confidence score and the source rows behind every field. Trade off. Every screen then has to show when the data is shaky and let people fix it, which is more to design and more to earn trust for.
02 16 RETURNED TO THE BUYER	Both sides negotiate blind Vendors cannot see where they stand, buyers can bluff about competing prices. Every extra round exists because someone is missing information.	WEIGHED AND DROPPED ✗ Sealed single round bidding, which removes negotiation ✗ Blind reverse auction, which keeps the asymmetry	THE CALL, AND WHAT IT COST A transparent reverse auction. Vendors see the standing best price but not each other's identity. Trade off. The buyer gives up bluffing leverage, and vendors who priced on opacity lose an advantage. Some will resist joining.
03 8 RETURNED TO THE BUYER	Clerical load crowds out strategy A buyer spends the day consolidating rather than thinking about category strategy, concentration or supplier risk.	WEIGHED AND DROPPED ✗ Full agent autonomy through to award, modelled at 8 minutes ✗ Assistive tips only, which leaves the 45 minutes intact	THE CALL, AND WHAT IT COST Agents own everything up to the award. The award and the signature stay human. Trade off. Three minutes of decision time stay in the cycle, so the outcome lands at 11 minutes rather than the theoretical 8.

THE REFRAME

Speeding up a bad process only gets you a fast bad process. Forty two of the forty five minutes are assembly work, so a slicker form was never the answer. It was to give the assembly to software and hand the buyer a decision that is already prepared. That moves the design question from **how do we digitize these nine steps** to **which steps should a human ever see**.

PRINCIPLES THAT CAME OUT OF IT

Bring the insight

If a user has to search for a finding, the system failed.

Assembly, never judgement

Agents prepare the decision. A person makes it and signs it.

Symmetric transparency

Both sides see the same market state, so neither guesses.

Everything traceable

Any number an agent shows opens to the rows behind it.

How the work ran, and what could not be verified

Business, engineering, the AI team and leadership all had a say. Each one wanted something different from the same design.

UNDERSTAND	ANALYSE	IDEATE	DESIGN	VALIDATE
Domain reading Process mapping Stakeholder interviews Glossary alignment	Task decomposition Time breakdown Journey maps Competitor teardown	Three directions Sketching Agent modelling Direction review	IA and navigation Wireframes Hi-fi screens Design system	Task-time walkthrough Engineering feasibility Leadership review Open questions

WORKING ACROSS FOUR GROUPS

Each group asked a different question of the same design. Getting the four answers to agree was most of the job.

Business	"Does this reduce cycle time and unit cost?" Drove the time-per-agreement target.
Engineering	"What can we actually ingest?" Forced the messy-data and low-confidence states into scope early.
AI / ML	"What is the agent allowed to decide?" Produced the autonomy levels on page 11.
Leadership	"Can I trust a number in a board deck?" Made traceability a first-class UI element, not a footnote.

STATED ASSUMPTIONS

- △ The 45-minute baseline comes from the brief, not from a timing study run here.
- △ The step-by-step breakdown was worked out from the documented process. It is a model, and the weakest link in the argument.
- △ Personas are composites built from stakeholder conversations and industry reading, not from interviews with buyers on site.
- △ Agent accuracy in the mockups is illustrative. Real thresholds would be set with the ML team against labelled data.
- ✓ The task-time walkthrough was run against the clickable prototype with five internal participants.

THE DISAGREEMENT THAT SHAPED THE PRODUCT

Business wanted autonomy

The fastest possible cycle is one where agents send RFPs, negotiate, and award without a human in the loop. On paper this hits eight minutes.

Leadership wanted signature

A committed price is a legal obligation. No CPO will let a model bind the company, however good the model gets.

Where it landed

Agents own everything up to the award; the award stays human. Costs three minutes, and it is the reason the product is deployable at all.

Most of the design work on this project was not drawing screens. It was finding the line between what a model may do and what a person must own, and then making that line visible in the interface.

Everyone digitizes the workflow. Nobody removes the work.

Five procurement products were pulled apart to see how much work each one still left with the user.

PRODUCT	STRENGTH	THE GAP DESIGNED AGAINST
Prokuria	Clean sourcing events and supplier management.	User still assembles the comparison and interprets it alone.
Procurify	Strong purchasing and approval controls.	Weighted to spend control after the fact, not to sourcing decisions.
Precoro	Approachable for mid-market teams.	Little category intelligence; no view of concentration or risk.
Fairmarkit	Automates tail-spend sourcing at volume.	Optimised for the long tail, not for renewing strategic agreements.
Freight negotiation tools	Proven that structured negotiation beats email.	Narrow to one category; no cross-category opportunity view.

CHOOSING A MARKET STRUCTURE

Replacing the email back and forth meant choosing an auction format. Four were studied, three ruled out.

Sealed-bid, single round Fast, but removes negotiation the buyers rely on.
Ascending (English) Wrong direction. One buyer, many sellers.
Blind reverse auction Keeps the information asymmetry that makes rounds drag.
Transparent reverse auction (chosen) Vendors see the standing best price, not each other's identity. Competition compresses rounds; the buyer cannot bluff, so vendors bid earnestly the first time.

WHAT THE RESEARCH SETTLED

Transparency is a speed feature Every negotiation round in the current process exists because someone lacks information. Publishing the best standing price removes rounds rather than adding pressure.
The vendor is a user, not a data source Competitors treat suppliers as recipients of forms. A vendor who cannot see where they stand replies slowly, which is the buyer's bottleneck too.
Intelligence is the open ground Category concentration, supplier risk and savings opportunities are reported by consultants, not surfaced by tools. That is where agents earn their place.

Three people, and three different reasons to distrust an agent

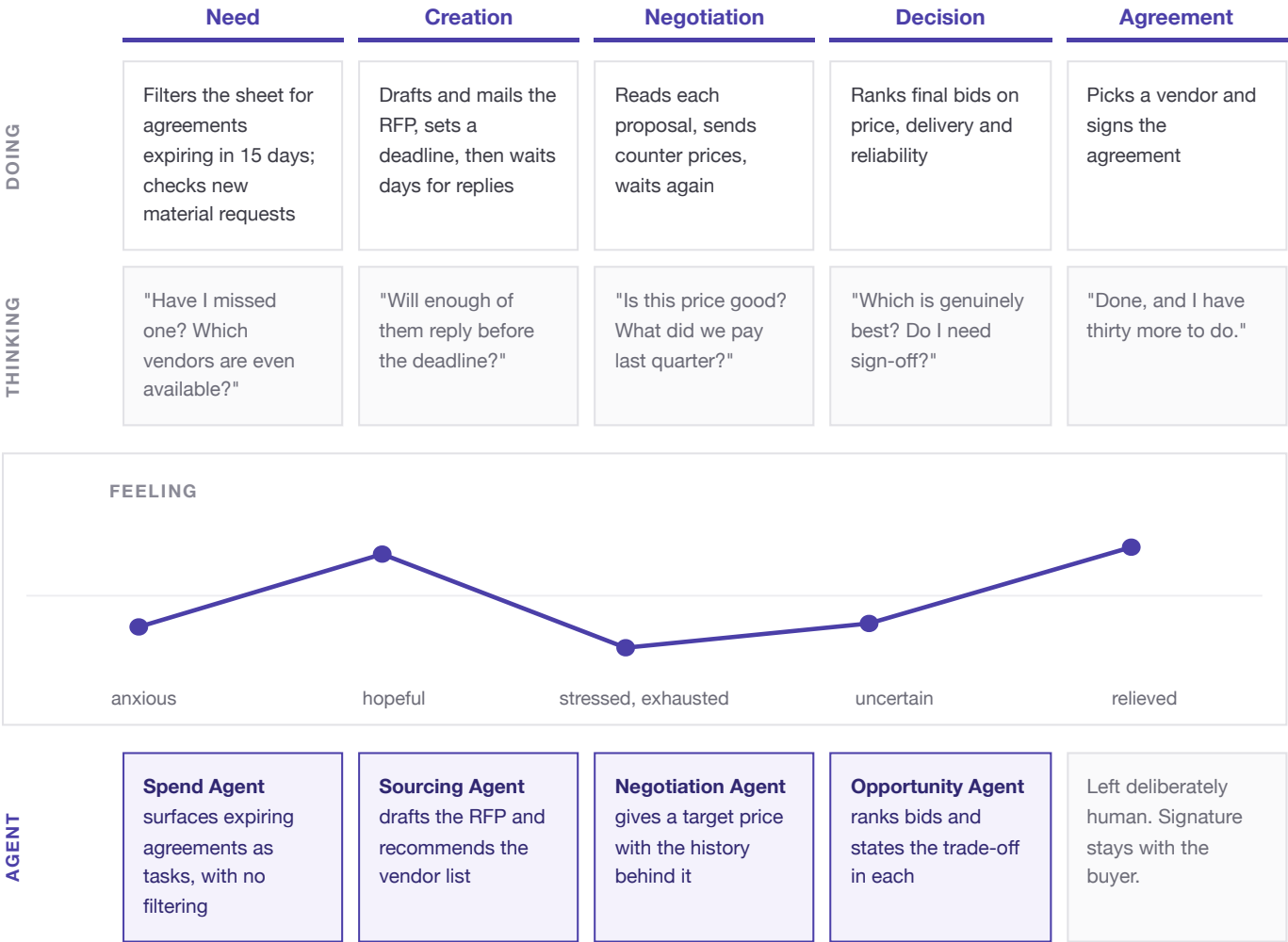
Composites from stakeholder conversations and industry reading. [△ modelled](#). Not validated with buyers at a customer site.

MR Meera R. Procurement Manager Pharma, 40 categories Primary user Daily, 3 to 5 hrs	GOAL Renew every expiring agreement on time at a defensible price, without living in her inbox.	FRUSTRATION Rebuilding the same comparison sheet from mail threads, then discovering a bid arrived after she started.	WHAT AN AGENT MUST EARN "Show me the rows behind the number." She will accept automation she can audit in one click.
RK Rakesh K. Vendor · sales owner Supplies 6 manufacturers Secondary user Bursty, deadline-driven	GOAL Win the tenders worth winning and not waste a week bidding on the ones he cannot.	FRUSTRATION Bidding blind. He learns he lost, never by how much, so his next bid is a guess again.	WHAT AN AGENT MUST EARN Neutrality. If the platform is the buyer's tool, he will keep negotiating off-platform.
DA Daniel A. Chief Procurement Officer \$48M annual spend Approver & buyer of the tool Weekly	GOAL Know where savings and supplier risk sit across the portfolio, in numbers he can defend upward.	FRUSTRATION Quarterly consultant decks that are stale on arrival and cannot be interrogated.	WHAT AN AGENT MUST EARN Where the number came from. A figure with no trail behind it is a liability in a board meeting.

The pattern across all three: nobody objected to automation. Everyone objected to automation they could not check. That is why confidence, source rows, and an approval gate appear on every agent surface in the product.

Procurement Manager, and where an agent should step in

One renewal, seen from the buyer’s desk, stage by stage.



WHAT THE MAP SHOWED

The low point is not at Decision, where the buyer does the skilled work. It is at Negotiation, where she is waiting on other people and cannot tell whether a price is good. Both problems are missing information, and both can be fixed without changing anyone’s job.

The relief at Agreement is also worth reading carefully. It is relief that a chore is over, not satisfaction at a decision well made, and she has thirty more waiting. A product that only makes the chore faster would leave that feeling exactly where it is.

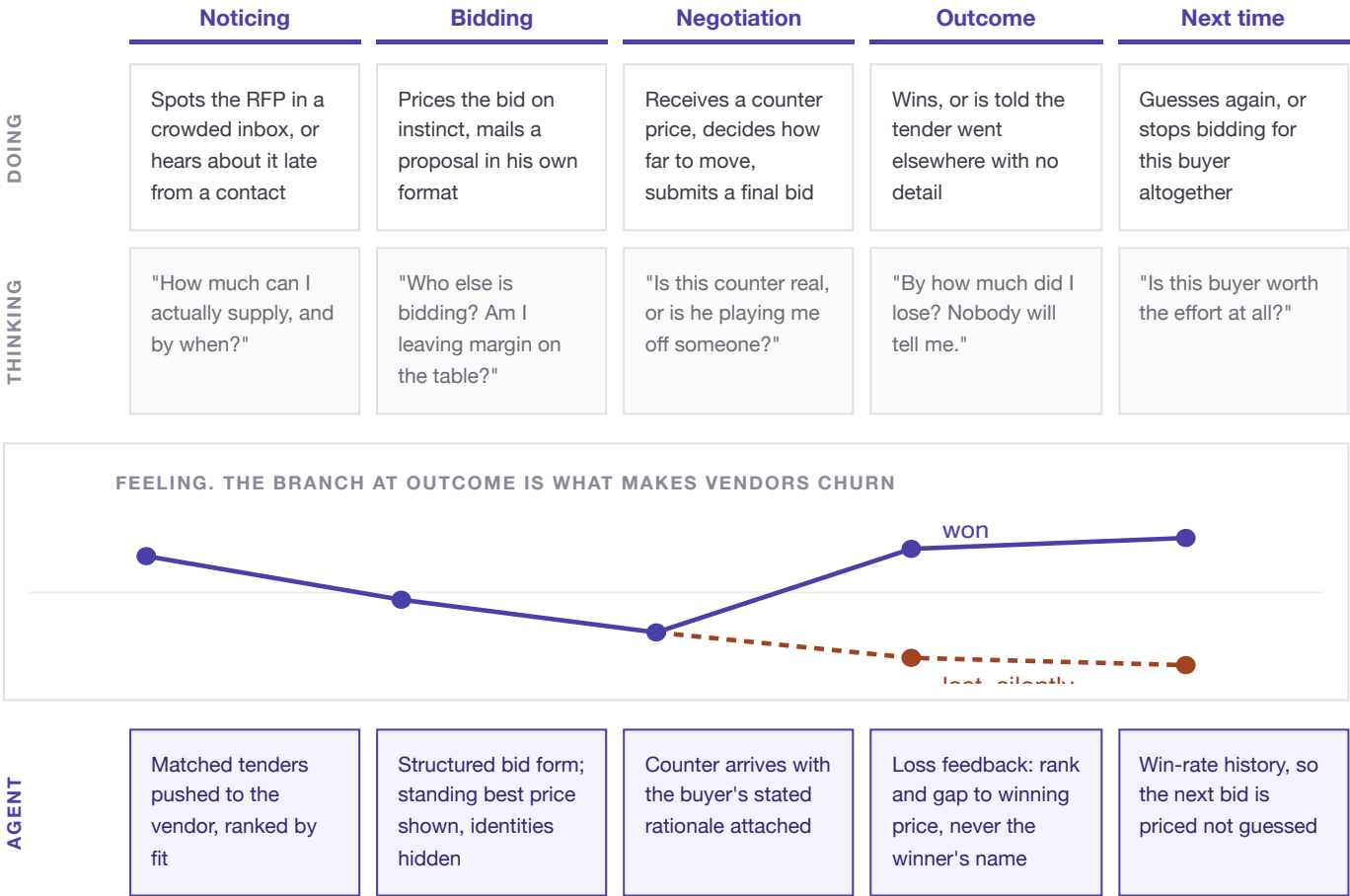
DESIGN CONSEQUENCE

Every agent in the row above intervenes **before** the stage where the buyer feels the pain, not during it. Prepared work removes the low; assistance offered mid-panic does not.

Agreement is the only stage with no agent. That absence is the product's promise: the machine hands over a decision, and a person makes it.

Vendor, the side most procurement tools forget

The buyer's slowest step is waiting for vendors. Fixing the vendor's experience is therefore a buyer feature, which is the argument that got vendor screens into scope.



THE HARDEST CALL IN THE PROJECT

How much to tell the loser

Silence is what makes vendors disengage, and a disengaged vendor pool is the buyer's problem within two cycles. But naming the winner turns loss feedback into competitive intelligence a supplier can use against the buyer.

Where it settled

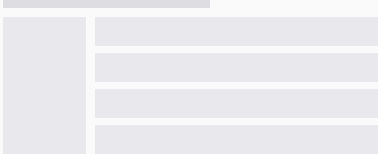
Rank and percentage gap to the winning price, never the winning vendor's identity. Enough to price the next bid rationally; not enough to reconstruct a competitor's cost base.

Still unresolved

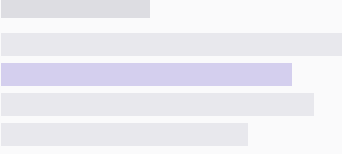
Whether transparency reads as fairness or as pressure. It is designed as fairness. Only research with vendors will settle it, and that research has not happened yet. It is the first item on page 16.

Three directions, judged against the forty-two minutes

Three ways to build it, scored against one question: how many of the forty-two clerical minutes does this remove?



DIRECTION A
Digitized workflow
Forms and tables replacing email. Everything the current process does, on one screen.
Removes ~14 of 42 minutes. The buyer still assembles and still interprets.



DIRECTION B
Auction marketplace
A live reverse auction as the centre of the product. Price competition does the work.
Great for negotiation, blind to strategy. Solves one stage of five.



DIRECTION C · CHOSEN
Agent command center
Agents watch the data continuously and hand the buyer prepared decisions. The auction sits inside it as one stage.
Removes ~34 of 42 minutes and absorbs B as a component.

SKETCHING THE SHIFT

The hard part of Direction C was not the layout, it was the front door. Early sketches opened on a wall of charts, which brought the original problem straight back: the user still had to go looking. Sketch four turned it around. The home screen became a queue of prepared decisions, and the charts moved behind them as evidence.



Placeholders. Drop the scanned ideation sketches here.

THE IDEA THAT WAS DROPPED

A chat-only interface

Sketch three replaced the whole product with a prompt box. It demoed beautifully and failed the persona test: Meera would have to already know what to ask, and Daniel could not audit an answer that arrived as prose.

Conversation survived as a way to **refine** an analysis, never as the way to find one. It lives in the custom-analysis path on page 13.

HOW THE DIRECTIONS WERE SCORED

CRITERION	WEIGHT	A WORKFLOW	B AUCTION	C AGENTS
Minutes removed from the cycle	High	Low	Medium	High
Works with messy first-load data	High	High	Medium	Medium
Auditable enough for a board number	High	Medium	Low	High
Buildable in one release cycle	Medium	High	Medium	Low

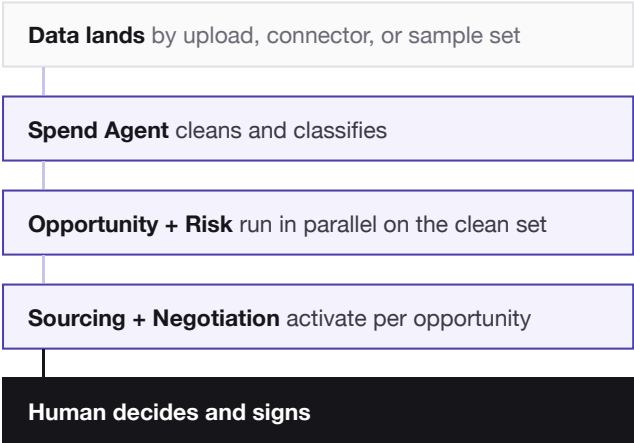
C lost on buildability and won on everything that mattered to the brief. Engineering agreed to it on the condition that the Spend Agent shipped alone first, which is why the empty and low-confidence states on page 15 are designed as a product, not as a stopgap.

Five narrow agents, one shared trust contract

One general assistant would have been easier to build and impossible to trust. Naming each agent after a single job makes its output checkable, and makes it obvious when an agent has no business answering.

AGENT	READS	PRODUCES	AUTONOMY
Spend Agent	Invoices, POs, GL lines	Normalised suppliers, category classification, concentration	Acts freely reversible, auditable
Opportunity Agent	Spend Agent output, price history	Ranked savings opportunities with modelled impact	Recommends human opens it
Sourcing Agent	Product master, vendor records, expiry dates	Draft RFP, recommended vendor list, cost model	Drafts human sends
Supplier Risk Agent	Delivery history, financials, concentration	Risk signals and changes, per supplier and portfolio-wide	Alerts never blocks
Negotiation Agent	Bid history, benchmarks, contract terms	Target price, leverage points, negotiation pack	Advises human negotiates

HOW THEY CHAIN



Nothing downstream runs on dirty data. If the Spend Agent's classification confidence drops below threshold, the chain pauses and asks. It does not guess and continue.

THE TRUST CONTRACT

Every agent output in the UI carries the same four things, in the same place. Consistency is what makes the automation legible.

- 1 **Attribution.** Which agent said this, marked with ♦.
- 2 **Confidence.** Stated as a band, not a false decimal.
- 3 **The trail.** One click to the transactions behind the figure.
- 4 **Reversal.** An obvious way to reject, correct, or teach.

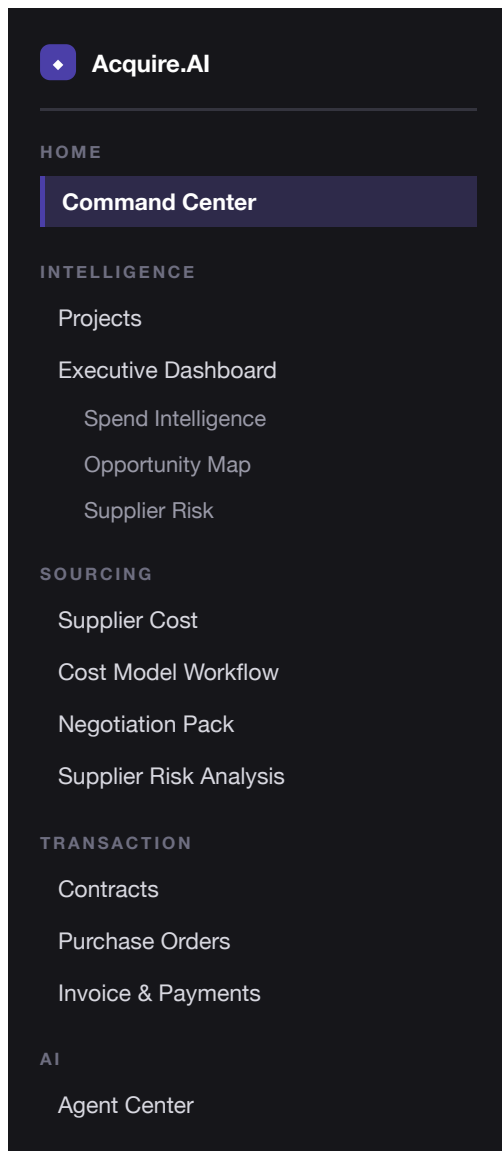
♦ Opportunity Agent · high confidence

\$1.2M in fragmented packaging spend

Across 14 suppliers · [2,146 transactions](#) · Dismiss

Navigation ordered by the procurement lifecycle, not by the org chart

The inherited structure grouped screens by the team that requested them. Regrouping by stage, from understand to discover to decide to act to monitor, meant a buyer could find a screen without knowing who built it.



THREE DECISIONS WORTH DEFENDING

Command Center is not a dashboard

It sits alone above Intelligence because it answers a different question. A dashboard reports state; the Command Center hands over work. Putting it inside Intelligence, as first proposed, would have buried it under charts.

Supplier Risk vs Supplier Risk Analysis

Two screens, nearly the same name, and the single biggest source of confusion in testing. They survive as separate items because they answer separate questions, so the labels carry the question, not just the noun.

Supplier Risk

Portfolio monitoring

"Which suppliers are risky?"

Supplier Risk Analysis

Single-supplier investigation

"Why is this one risky, and what do I do?"

Agent Center at the bottom, on purpose

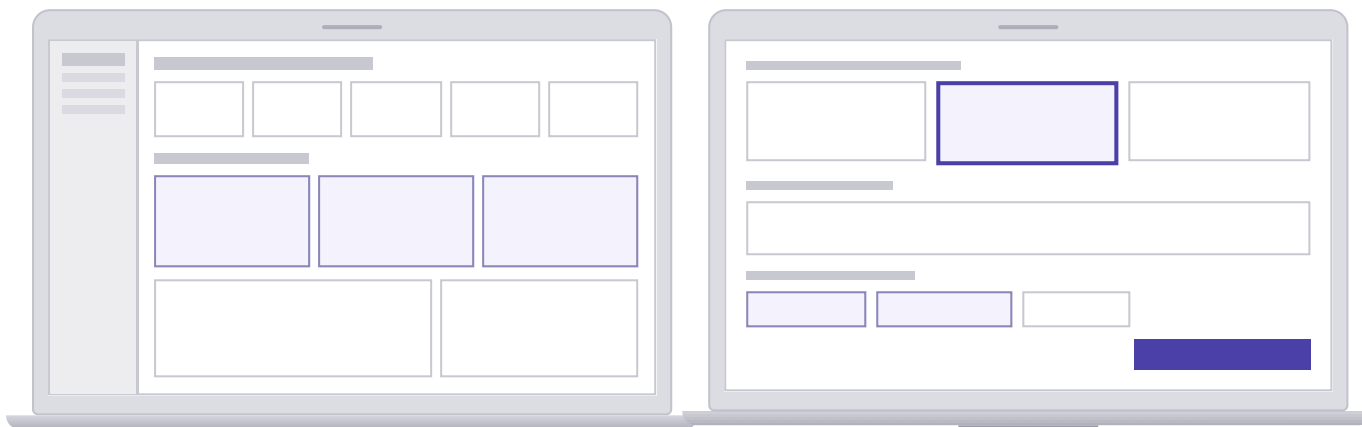
Agents are how the product works, not a place you visit. The Center exists for configuration and audit; the agents themselves appear inline wherever their output is relevant. Nobody should have to go to the AI section to get AI.

PROGRESSIVE DISCLOSURE

A first-time user sees Command Center and Projects only. Sections unlock as the data behind them arrives, so the sidebar grows with the account instead of presenting sixteen dead ends on day one.

Structure first, in grey boxes, so the idea stayed cheap to argue with

Layout only, no colour and no real numbers, walked through with the team before anything was made to look finished.



Command Center

KPIs compress to a single strip. The three tinted cards are the prepared decisions, the only thing above the fold that is not a number.

Create project

Analysis type, then data, then agents. Agents arrive pre selected with a reason, so the user edits a recommendation instead of building one.



Analysis in progress

Not a spinner. Per-agent lines with real counts, because a wait that explains itself reads as competence rather than delay.

Project result

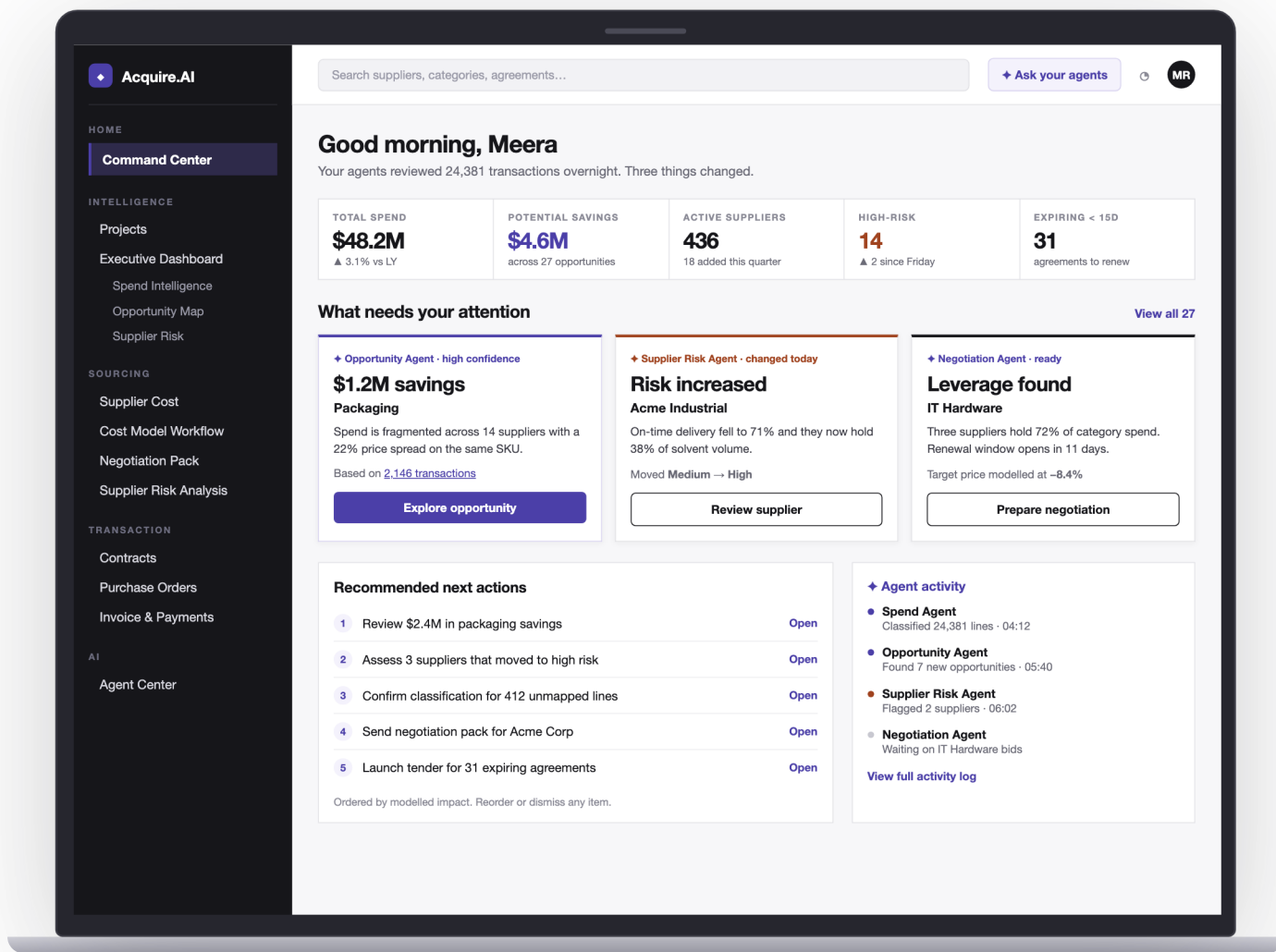
Findings on top, tabs below. Every finding states what it means, the impact, and the action. A chart alone was never enough.

WHAT THE WIREFRAME ROUND CHANGED

Two things broke in the walkthrough. Participants read the KPI strip as the main content and missed the recommendation cards entirely, so the cards grew and the KPIs shrank to a single row. And nobody understood **"Activate agents"**. They expected a setting, not a start. It became **"Run analysis"** everywhere it appeared outside onboarding.

Command Center, the queue of prepared decisions

This is the screen everything else leads to: the first thing a buyer sees each morning.



Numbers, then decisions

The KPI strip is deliberately one row of small type. It orients; it is not the work.

◆ marks every agent claim

One glyph, used nowhere else, so a user always knows which statements are inferred.

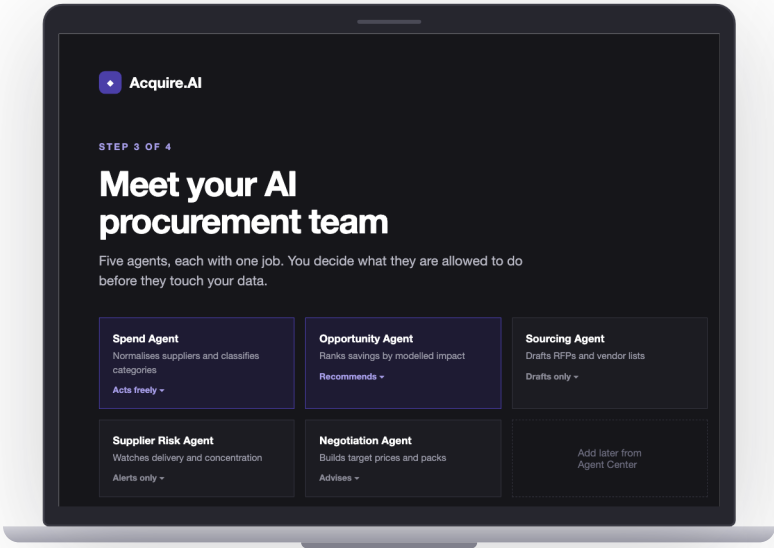
One primary action per card

Filled for the highest-impact item only. Everything else is outlined, so the queue has a head.

The subhead does the heaviest lifting on this screen. "Your agents reviewed 24,381 transactions overnight. Three things changed." answers what changed, what matters, and how much work was done for you, in one line, before the user has parsed a single number.

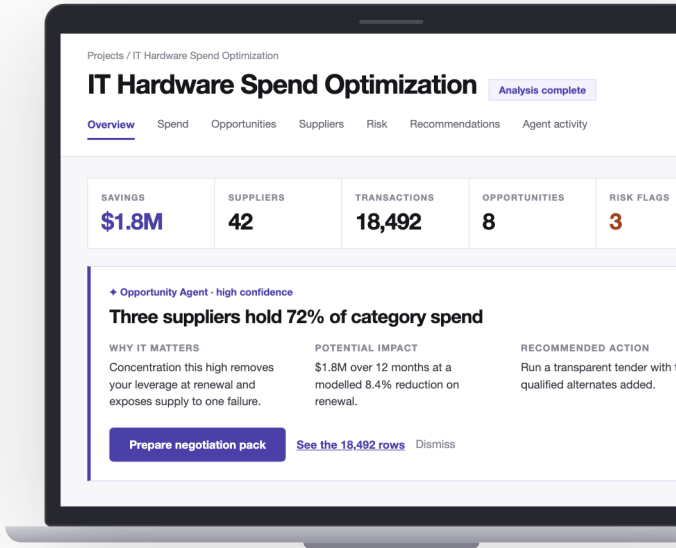
Onboarding sets the contract. The workspace keeps it.

Two screens from either end of the story: the first five minutes in the product, and a finished piece of work.



Onboarding · agent activation

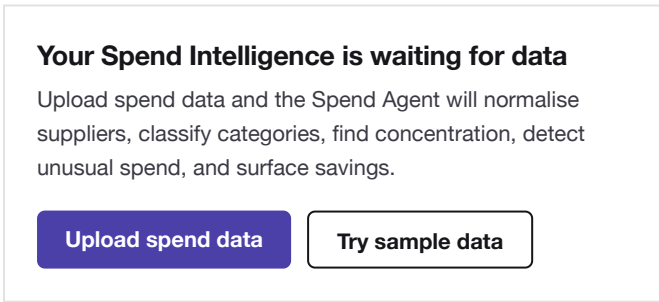
Autonomy is set here, before any data moves, and every level is a dropdown the user can lower later. Consent up front is what buys latitude afterwards.



Project workspace · result

Every finding is written as three columns, why it matters, potential impact, recommended action, so the output is a decision brief, not a chart that must interpret.

THE EMPTY STATE CARRIES THE PROMISE



"No data available" tells a user they failed. Listing what the agent will do turns the same screen into the strongest sales argument in the product.

MICROCOPY RULES THE SYSTEM RUNS ON

Loading...	Opportunity Agent is analysing your spend
No opportunities found	No high-impact opportunities identified yet
Supplier has risk	Supplier risk increased, review recommended
Click here to see your data	Explore your spend

Every state names the agent doing the work and what it is doing. Waiting is tolerable when it is legible.

Responsive behaviour

On tablet the sidebar collapses to icons and the recommendation row becomes two columns. On mobile it inverts entirely: alerts, then recommendations, then KPIs. A buyer on a phone is reacting to something, not browsing.

Restraint on AI styling

Gradients and glows appear only on agent surfaces, and only as a tinted background. Nothing pulses, nothing animates on load. In an enterprise tool, visual excitement around a number reads as marketing, which undermines the number.

One accent, four greys

The palette holds a single purple, one warning ochre, and a charcoal-to-hairline grey ramp. Colour is reserved for meaning, whether agent, risk or action, so nothing decorative competes with a status.

Eleven minutes, and only the judgement is left

One renewal, start to finish, timed on the clickable prototype with five internal participants. The brief asked for 15 minutes.



WHAT WOULD BE DONE DIFFERENTLY

The baseline was modelled, not measured

The 42-of-45-minutes argument carried the whole project, and it rests on a breakdown made on paper. Two hours of watching one buyer work would have turned it into evidence.

The happy path came first

Low-confidence and conflicting-agent states arrived late, after engineering raised them. In an agentic product the uncertain state is the normal state, and it should have been screen one.

Vendors were barely tested with

Half the platform's value depends on suppliers adopting it, and every participant in the walkthrough was a buyer.

NEXT

Instrument the agreement cycle

Replace the modelled baseline with real per-stage timing from the pilot.

Design the disagreement state

What the UI does when the Risk Agent and Opportunity Agent recommend opposite things about the same supplier.

Vendor-side research round

Test whether transparent bidding reads as fair or as pressure. If it reads as pressure, adoption stalls and the speed gain evaporates.

Correction as a teaching loop

Today a user can reject a classification. Next, that rejection should visibly improve the agent.